

Technical Document #231: Natural Language Processing - Comprehensive Overview

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Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Text summarization generates a concise summary of a longer text. Extractive summarization selects important sentences while abstractive summarization generates new text.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Question answering systems extract answers from a given context.
- Text summarization generates a concise summary of a longer text.
- Machine translation converts text from one language to another.